

Cs301

Midterm important questions 2024

VU360

Lecture 1-5:

1. What is the definition of data structures and their importance?
2. What are the different types of data structures (arrays, linked lists, stacks, etc.)?
3. How do you implement and use arrays in programming?
4. What is the concept of dynamic memory allocation and its importance?
5. How do you use pointers and references in programming?

Lecture 6-10:

6. What is the concept of linked lists and their implementation?
7. How do you implement and use stacks and queues?
8. What is the concept of trees and their implementation?
9. How do you implement and use graphs in programming?
10. What is the concept of hashing and its importance?

Lecture 11-15:

11. How do you implement and use sorting algorithms (Bubble sort, Selection sort, etc.)?
12. What is the concept of searching algorithms and their importance?
13. How do you implement and use recursive algorithms?
14. What is the concept of backtracking and its importance?
15. How do you implement and use greedy algorithms?

Lecture 16-20:

16. What is the concept of dynamic programming and its importance?
17. How do you implement and use memoization and tabulation?
18. What is the concept of bit manipulation and its importance?
19. How do you implement and use bitwise operations?

20. What is the concept of string manipulation and its importance?

Lecture 21-22:

21. How do you implement and use advanced data structures (trie, suffix tree, etc.)?

22. What is the concept of data structure optimization and its importance?